

Microscopy Image Viewer

Developer Specification v1.0

1. Project Goal

Develop a native Lazarus/Free Pascal image viewer optimized for rapid inspection and sorting of scientific microscopy images.

The software shall emphasize:

- Immediate response to user actions.
- Minimal user interface.
- Mouse-driven navigation.
- Recursive directory browsing.
- High-performance image preloading.
- Modular architecture.

The viewer is **not** intended to be a photo management application.

2. Design Principles

Performance First

Every architectural decision should improve browsing speed.

If a feature slows image navigation, it should be redesigned.

Modular Design

The application consists of independent subsystems.

No subsystem should know unnecessary details about another subsystem.

Command-Based Architecture

Every user action produces a command.

Examples:

NextImage

PreviousImage

NextDirectory

PreviousDirectory

ZoomIn

ZoomOut

FitToScreen

OriginalSize

Exit

Future additions:

MoveImage

CopyImage

DeleteImage

Magnifier

The renderer should never care whether a command came from:

- mouse wheel
 - mouse gesture
 - keyboard
 - scripting engine
-

3. Major Modules

TNavigator

Responsibilities

- Scan directory tree
- Build directory list
- Build image lists
- Maintain current position
- Wrap navigation
- Provide filenames

Public interface

OpenRoot()

CurrentFilename()

NextImage()

PreviousImage()

NextDirectory()

PreviousDirectory()

ImageCount()

DirectoryCount()

CurrentDirectoryName()

No rendering code.

No image decoding.

TImageLoader

Responsibilities

- Load supported formats

Supported formats

TIFF

JPEG

PNG

GIF

BMP

Future

WebP

AVIF

JPEG XL

TRenderer

Responsibilities

Display already-loaded images.

Functions

ShowBitmap()

Zoom()

FitToScreen()

OriginalSize()

Fullscreen()

Background color

Interpolation mode

The renderer never opens files.

TPreloader

Runs in a background thread.

Responsibilities

Predict upcoming images.

Decode them.

Store decoded bitmaps.

Maintain cache.

The cache should contain several images before and after the current image.

Later versions may bias the cache according to browsing direction.

TMouseEngine

Responsibilities

Track

Mouse position

Movement

Wheel

Buttons

Determine

Current quadrant

Gesture

Generate commands.

No viewer logic.

TThumbnailCache

Responsibilities

Generate thumbnails.

Load thumbnails.

Save thumbnails.

Detect changed files.

No SQL database.

Simple filesystem cache.

4. Navigation Model

The scanner builds an internal structure.

Root

↓

Directory List

↓

Image List

Each directory contains

Directory Name

Image List

Current Image Index

Navigator maintains

Current Directory Index

Current Image Index

Navigation never accesses the filesystem directly after the initial scan.

5. Rendering Pipeline

Navigator



Filename



ImageLoader



Bitmap



Renderer



Screen

The renderer never receives filenames.

Only decoded images.

6. Mouse System

The screen is divided into four quadrants.

Each quadrant owns an independent command map.

Example

Quadrant 1

WheelUp

→ ZoomIn

WheelDown

→ ZoomOut

Quadrant 2

WheelUp

→ NextImage

WheelDown

→ PreviousImage

Gestures

Down

Left

Right

Up

Diagonal movements may be added later.

7. Configuration

Version 1

INI file.

Later

Lightweight scripting language.

Possible syntax

on WheelUp Quadrant1 ZoomIn

on GestureDown Exit

The scripting engine should translate script into commands.

8. Sorting System

Future version.

Screen edge activates destination popup.

Example

Good

Interesting

Reject

Measurements

Archive

Selecting an item performs

Move

or

Copy

without interrupting navigation.

9. Coding Standards

One class per unit.

Clear interfaces.

Minimal global variables.

No GUI code inside engine classes.

Business logic separated from presentation.

Thread-safe communication.

Comment public interfaces.

Avoid premature optimization except inside the preload engine.

10. Version 1 Feature Set

Mandatory

- ✓ Recursive directory scan
- ✓ TIFF support
- ✓ JPEG support
- ✓ PNG support
- ✓ GIF support
- ✓ BMP support
- ✓ Fullscreen
- ✓ Fit to screen
- ✓ Original size
- ✓ Next image
- ✓ Previous image
- ✓ Next directory
- ✓ Previous directory
- ✓ Wrap-around navigation
- ✓ Background preload thread
- ✓ Mouse wheel navigation
- ✓ Mouse gesture exit

✓ Quadrant-aware mouse commands

Optional

Thumbnail cache

Animated GIF playback

Magnifier

11. Future Features

- Multipage TIFF navigation
 - 16-bit grayscale display
 - Scientific LUTs
 - Histogram display
 - ROI selection
 - Measurement tools
 - User scripting language
 - Plugin architecture
 - Dual-monitor mode
 - Contact sheet view
 - Persistent thumbnail cache
 - Edge-based sorting popup
 - Drag-and-drop folder opening
-

Guiding Principle

The application exists for one purpose:

Allow a scientist to move from one image to the next with virtually no delay while keeping their hands on the mouse.

Every new feature should support that workflow. If it does not, it should be questioned before implementation.